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(57) Abstract:

This invention is a smart system for ambulances that does two vital jobs at once: monitoring the patient and clearing traffic. The health part uses sensors (for ECG, oxygen, blood pressure, and stress) connected to a microcontroller (like Arduino Mega) to collect vital signs. It cleans up the signals, packages the data, and sends it via Wi-Fi to a hospital cloud page for doctors to see in real-time. At the same time, the traffic part uses a separate microcontroller (like Arduino Nano) with GPS. When the ambulance gets near a traffic light, it calculates the distance and automatically sends a radio command (using an nRF24L01+ chip) to that junction. A receiver box at the light gets the command, checks it, and safely overrides the normal lights to turn green for the ambulance, creating a clear path. Both parts work independently off the ambulance's 12V power, cutting down both diagnosis time and travel time to save lives.

FORM 2

THE PATENTS ACT, 1970

(39 OF 1970) &

THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

TITLE OF THE INVENTION

AN INTEGRATED SYSTEM FOR REAL-TIME PATIENT HEALTH MONITORING AND INTELLIGENT TRAFFIC SIGNAL CONTROL FOR EMERGENCY AMBULANCES

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PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be perform.

FIELD OF THE INVENTION

[0001] This invention is basically about making ambulances smarter. It sits at the crossroads of a few key areas: emergency medical services (EMS), telemedicine, embedded electronics, and smart traffic systems. More plainly, it's a full hardware and software setup built right into a regular road ambulance.

[0002] The main idea is to tackle two big, usually separate, problems at the same time. First, we continuously check on the patient's critical health signs and send that information ahead to the hospital. Second, the system automatically finds and clears a path through traffic by talking to upcoming traffic signals.

[0003] By doing both jobs together, the system aims to cut down the total "time-to-treatment" in an emergency. It fights the information delay (doctors waiting for the patient) and the travel delay (the ambulance stuck in traffic) all at once.

BACKGROUND OF THE INVENTION

[0004] In a serious medical emergency, two things are most important: getting the patient to the hospital fast, and getting information about the patient to the doctors even faster. In crowded cities today, both of these are major challenges.

[0005] Looking at patient monitoring first: While modern ambulances do carry equipment like ECG and oxygen monitors, these are often standalone devices. The data they collect mostly stays inside the ambulance, sometimes just written down on paper. This creates a kind of "information blackout" between the accident site and the hospital ER. Doctors can't start planning proper treatment until the patient physically arrives and gets hooked up to hospital machines, wasting those first critical "golden hour" minutes.

[0006] Some newer ideas use mobile networks (3G/4G) to send data. But these can be slow, unreliable in tunnels or during network rush hour, and might fail in a disaster. Also, they often only send one type of data, like just the heart signal, instead of a complete set of vital signs.

[0007] Now, about traffic clearance: Ambulances today depend on loud sirens and flashing lights. In heavy, gridlocked traffic, this often doesn't work because other cars simply have no space to move. Some tech solutions have been tried:

- **Infrared (IR) systems:** These need a clear line-of-sight from the ambulance to a receiver on the signal post. Their range is short (under 100m), doesn't work around corners, and can fail in bad weather.
- **RFID tags:** This would need special readers at every single junction and tags on every ambulance. The cost of setting this up across a whole city is far too high.
- **Central cellular systems:** Here, the ambulance sends its location via mobile data to a city traffic server, which then tries to change the signals. This introduces many delays—network lag, server processing time, and signal time to the junction—and isn't a guaranteed, instant solution.

[0008] The core issue with current approaches is that they treat "health monitoring" and "traffic clearance" as totally separate problems. Research papers and products usually solve one or the other. A smart health monitor for ambulances doesn't fix traffic jams. A fancy traffic pre-emption system doesn't give doctors any patient data. This misses the bigger picture: the total emergency time is the sum of travel time and hospital preparation time. Just improving one part doesn't give the best result.

[0009] Also, ideas that try to combine both are often too complex, expensive, and power-hungry, using heavy computers and special modems. This makes them impractical for fitting into ordinary ambulances, especially in places where budgets are tight.

[0010] Because of all this, there's a real and ongoing need for a single, affordable, reliable system that does two things well: provides solid, multi-parameter patient data and guarantees a clear road, all while running from a standard ambulance battery using common, modular parts.

OBJECTS OF THE INVENTION

[0011] The main goal of this invention is to get past the limitations mentioned above. We want to create one unit that physically and functionally combines patient health tracking with a system to clear traffic.

[0012] Another aim is to design a health tracking part that can read, clean up, and process signals from multiple medical sensors at the same time—like a light-based pulse oximeter for oxygen levels, a sensitive ECG amplifier for heart electrical activity, a pressure cuff for blood pressure, and a simple circuit to measure skin sweat response (GSR).

[0013] A further object is to run real-time software filters on a microcontroller to clean up the raw sensor data, for example, to remove electrical hum (50/60 Hz) and slow drift from the ECG signal before sending it out.

[0014] We also want to set up a strong, fast link to send this data from the moving ambulance to the hospital. This involves using a common Wi-Fi module to connect to a mobile hotspot and a simple internet protocol to send neat data packets (like in JSON format) to a hospital computer screen.

[0015] For the traffic side, a key object is to have a separate microcontroller constantly reading location data from a standard GPS module.

[0016] This microcontroller should run a smart "geofencing" program. This program calculates the straight-line distance (using something like the Haversine formula for the Earth's curve) between the ambulance and known traffic junctions, and kicks into action when the ambulance gets close (say, within 500-1000 meters).

[0017] When it gets close enough, the system should create a small, secure digital message. This message pinpoints the target junction and tells it which way the ambulance is coming from (North, East, etc.), figured out from the GPS direction.

[0018] This message must be sent using a dedicated, low-cost radio chip (like the 2.4 GHz nRF24L01) that works independently of phone networks, on its own channel.

[0019] Another important piece is the receiver box at the traffic junction. This unit must listen for the ambulance's radio message, check it's genuine, and then safely take over the normal traffic light controller to run a special "green corridor" sequence.

[0020] The whole system should be designed so the health part and the traffic part work side-by-side on their own, sharing just the ambulance's power. If one part has a problem, the other should keep working.

[0021] Finally, we want to build all this using easy-to-find, modular parts like common microcontroller boards (Arduino style) and standard electronics to keep cost low, maintenance simple, and future changes easy.

SUMMARY OF THE INVENTION

[0022] This invention describes a unified system and method that tackles the twin problems of patient data blackout and ambulance traffic delay head-on, together.

[0023] In simple terms, the invention is an integrated system for an ambulance. It has two main parts living in the same box: a Health Monitoring Module (HMM) and a Traffic Clearance Module (TCM), both powered by the ambulance's own electrical system.

[0024] The HMM is like a multi-channel data collection hub. It connects a bunch of sensors—a MAX30100 chip for pulse and oxygen (using I^2C), an AD8232 chip for ECG (analog output), a pressure sensor for blood pressure, and a simple circuit for skin conductance (GSR)—to a main microcontroller, like an Arduino Mega. This microcontroller reads the signals, runs calculations and filters (like cleaning up the ECG waveform), figures out things like heart rate, and bundles the data. A small OLED screen shows the info locally. Then, an ESP8266 Wi-Fi chip sends this data pack to a cloud service (like Blynk), where hospital staff can see live graphs and numbers on a dashboard.

[0025] The TCM is the "pathfinder." It's built around a smaller controller, like an Arduino Nano, hooked to a GPS module. This Nano constantly checks where the ambulance is. It has a list of important junctions. Using a bit of math (the Haversine formula), it calculates the distance to the next junction. When it gets within a set range, it creates a short radio command saying "Junction X, I'm coming from the East." This command is broadcast by an nRF24L01+ radio chip.

[0026] At the traffic junction, there's a matching receiver box (the JCU). It has the same kind of radio, listening. When it gets the right command for its junction, a microcontroller inside (like an Arduino Uno) checks it and then takes control. It safely changes the traffic lights—usually turning the current greens to yellow, then red, and finally giving a solid green to the ambulance's lane for enough time to pass, before going back to normal.

[0027] The real innovation here is that these two modules—the HMM sending patient data and the TCM clearing the road—work completely at the same time, independently. They don't wait for each other. This parallel operation, packed into one affordable system, attacks both parts of the emergency delay simultaneously, which is a big step up from older, separate approaches.

BRIEF DESCRIPTION OF DRAWINGS

[0028] To help explain the invention clearly, we've included several drawings. Please note these are just to illustrate the ideas and shouldn't limit what the invention covers.

[0029] FIG. 1 shows the big picture—a block diagram of how all the main parts of the smart ambulance system (100) fit together, including power and data flow.

[0030] FIG. 2 zooms in on the Health Monitoring Module (HMM, 200). It's a block diagram showing the group of sensors (202)—pulse meter (210), ECG

(212), blood pressure (214), GSR (216)—and how they link to the main controller (204), the local screen (206), and the Wi-Fi unit (208).

[0031] FIG. 3 is a similar block diagram for the Traffic Clearance Module in the ambulance (TCM-A, 300). It shows the GPS (302), its controller (304), and the radio chip (306).

[0032] FIG. 4 shows the other side: the Junction Control Unit (JCU, 400) block diagram. It has the receiving radio (402), its own controller (404), and how it connects to the actual traffic light controller (406).

[0033] FIG. 5 is a flowchart (500) that maps out the two main processes happening at once: the continuous loop of health monitoring (510) and the step-by-step sequence for traffic clearance (520-560).

[0034] FIG. 6 is a circuit diagram for how the Pulse Oximetry sensor (210) is wired to the main controller (204).

[0035] FIG. 7 is a circuit diagram for wiring the ECG sensor (212) to the controller (204).

[0036] FIG. 8 is a circuit diagram for the Blood Pressure sensor (214) connections.

[0037] FIG. 9 is a circuit diagram for the GSR sensor (216) connections.

[0038] FIG. 10 is a wiring diagram showing how the OLED display unit (206) connects to the main controller (204).

[0039] FIG. 11 is a wiring diagram for connecting the GPS module (302) to its controller (304).

[0040] FIG. 12 is a wiring diagram for the Wi-Fi module (208) connection.

[0041] FIG. 13 is a wiring diagram detailing the SPI pins for connecting the nRF24L01+ radio chip (306).

[0042] FIG. 14 is a photograph of our physical ambulance unit prototype with the system installed.

[0043] FIG. 15 is a photograph of our physical traffic signal controller prototype.

[0044] FIG. 16 is a photograph showing the whole setup together—the ambulance prototype and the traffic controller prototype.

DETAILED DESCRIPTION OF THE INVENTION

[0045] Let's get into the details of how we built and how the system works. This description should give someone with skills in electronics and programming enough to recreate it.

[0046] **System Overview and Power (Looking at FIG. 1):** We designed the whole system (100) as a kit that can be added to existing ambulances. It starts by tapping into the ambulance's standard 12V DC power (102), which you can find at an accessory socket or the fuse box. Vehicle power is messy, with voltage spikes and drops, so we added a power conditioning stage (104).

[0047] This stage has two steps. First, a protection diode clamps any dangerous spikes above 15V. A big capacitor (like 470µF) smooths out the input. Then, an LM2596 switching regulator efficiently steps the voltage down to a steady +5.0V. From that, a smaller AMS1117-3.3 linear regulator gives us a very clean +3.3V line for the sensitive sensor and communication chips.

[0048] For the brains, we didn't use one super-chip. We used two common microcontroller boards that talk to each other. An Arduino Mega 2560 acts as the main controller for the health module (it's the "first microcontroller" (204)). An

Arduino Nano handles the traffic clearance jobs (the "second microcontroller" (304)). We connected them with a simple serial link (UART) between spare pins (Mega's pin 14 & 15 to Nano's pin 0 & 1) so they can share status updates, but they can work fine on their own if this link fails.

[0049] Deep Dive into the Health Module (Looking at FIGS. 2, 6, 7, 8, 9): The Health Monitoring Module (HMM, 200) is where we gather all the patient data. As you can see in FIG. 2, the sensor group (202) has four main pieces:

- **The Pulse Oximeter (210):** We used a MAX30100 chip. It's powered by 3.3V. We connected its SDA and SCL pins (the I²C bus) to the Mega's dedicated I²C pins (20 and 21), with 4.7kΩ pull-up resistors, as shown in FIG. 6.
- **The ECG Sensor (212):** This is an AD8232 board. It also needs 3.3V. Its main OUTPUT pin goes to the Mega's Analog Pin A0 (see FIG. 7). Its LO+ and LO- pins (for detecting if electrodes fall off) go to Digital Pins 10 and 11. We keep its shutdown pin low to keep it always on.
- **Blood Pressure (214):** This part (FIG. 8) has a small air pump, a release valve, and a pressure sensor. We control the pump and valve with a transistor switched by Digital Pin 5. The sensor's output wire goes to Analog Pin A1.
- **GSR Sensor (216):** This is simple (FIG. 9). We use two metal electrodes for the fingers. A voltage divider gives a constant 0.5V across them. The changing voltage across a 10kΩ resistor in series, which depends on how sweaty the skin is, is read at Analog Pin A2.

[0050] The Mega (204) runs a program that does several things in a loop. It sets up the ADC to read fast, starts the I²C bus, and opens serial ports for the Wi-Fi and the link to the Nano.

[0051] For the ECG signal from pin A0, we sample it 200 times a second. Right in the code, we run a digital bandpass filter (a 2nd order IIR filter) to block very slow drifts and signals above 40 Hz (like muscle noise). We also look at this cleaned-up signal to find the sharp R-peaks and calculate the heart rate.

[0052] Every second, we pack all the readings into a neat text format called JSON. It looks something like this:

`{"id":"AMB001","t":1677672000,"ecg":[325,328,330,...],"spo2":98,"bpm":72,"sys":118,"dia":76,"gsr":512}`

[0053] This string is sent out of the Mega's Serial1 port to the ESP8266 Wi-Fi module (208). We programmed the ESP8266 to connect to a smartphone's hotspot and then send this data securely to the Blynk cloud. The hospital can then pull up the Blynk app and see the patient's ECG wave and vital signs updating live.

[0054] How the Traffic Clearance Module Works (Looking at FIG. 3): The Traffic Module (TCM-A, 300) is all about location and timing. The GPS module (302) (a NEO-6M) constantly spits out location sentences. The Nano (304) reads this via a software serial link on its pins 2 and 3.

[0055] In the Nano's code, we store the latitude and longitude of key traffic junctions. Every few seconds, it grabs the ambulance's current location from the GPS. Then it uses the Haversine formula—which accounts for the Earth's round shape—to calculate the straight-line distance to each junction.

[0056] If the ambulance is moving toward a junction and comes within, for example, 800 meters of it, the system triggers.

[0057] It then figures out which way it's heading relative to the junction (like "approaching from the East") and creates a small 6-byte radio packet. This packet has a start code, the junction's ID number, the direction code, a sequence number, and a checksum for error-checking.

[0058] This packet is handed to the nRF24L01+ radio chip (306). The Nano talks to this chip over SPI (pins 11, 12, 13) and uses two other pins (9 and 10) for control. We set the radio to a specific channel (like 76) and a high speed (2 Mbps). Then it starts broadcasting the packet over and over as it approaches.

[0059] At the Junction – Taking Over the Lights (Looking at FIG. 4): Sitting at the traffic junction is the Junction Control Unit (JCU, 400). It has the same model of nRF24L01+ radio (402), but set to listen mode. Its brain is an Arduino Uno (404).

[0060] Normally, the Uno just runs the standard traffic light pattern. But it's always listening. When a radio packet comes in, it checks three things: 1) Is the start code correct? 2) Does the checksum match (meaning no corruption)? 3) Is the junction ID in the packet meant for this junction?

[0061] If all three are good, it starts an override routine. Let's say the command was "East." The Uno will first change the North-South lights to yellow, then red, making sure those lanes stop. After a brief all-red safety pause, it switches the East light to solid green. It holds this green for a pre-set time (say, 30 seconds) or until it gets a "clear" signal. Then it safely returns the lights to their normal automatic cycle.

[0062] Connecting the Modules (Looking at FIGS. 10-13): To make everything work together, getting the wiring right is key. The diagrams from FIG. 10 to FIG. 13 show exactly how we connected the key parts:

- **FIG. 10** shows the 4-pin cable for the OLED screen (206), connecting VCC, GND, SDA, and SCL.
- **FIG. 11** shows the 4-pin connection for the GPS (302) – VCC, GND, TX, RX – to the Nano.
- **FIG. 12** shows how the ESP8266 Wi-Fi module (208) connects with VCC, GND, TX, RX to the Mega.
- **FIG. 13** is crucial for the radio link. It details all 8 connections for the nRF24L01+ (306): power (3.3V, GND), the SPI pins (MOSI, MISO, SCK), and the control pins (CE, CSN).

[0063] The Physical Prototype (Looking at FIGS. 14-16): We didn't just design it on paper. We built it. FIG. 14 is a photo of our prototype unit installed in a model ambulance setup. FIG. 15 shows our standalone traffic junction controller box. FIG. 16 shows both units working together, demonstrating the complete system idea.

[0064] The Core Concept – Parallel Operation (Looking at FIG. 5): If you follow the flowchart in FIG. 5, you see the magic. On the left, the Health Monitoring loop (510) runs non-stop: Sense -> Process -> Show -> Send. On the right, the Traffic Clearance process (520-560) runs on its own, triggered by location. They don't depend on each other to start. This parallelism means the hospital is getting ready **while** the road is being cleared, which is the key to saving crucial minutes.

[0065] Other Ways to Build It: We used specific Arduino boards to keep it simple and cheap. But the idea isn't locked to these. You could use a more powerful chip like an ESP32 to run both modules. You could swap the nRF radio for a LoRa module for longer range. The health data could go over a 4G network instead of Wi-Fi. The traffic junction box could be part of a bigger city traffic network. The core invention is the **combined, parallel operation** of the two functions in one system.

CLAIMS

We claim:

1. An integrated emergency response system for a vehicle, comprising:
   (a) a health monitoring module configured to acquire physiological data from a patient within said vehicle;
   (b) a traffic clearance module configured to communicate with external traffic control infrastructure;
   (c) a common power supply unit operatively coupled to both said health monitoring module and said traffic clearance module;

   wherein said health monitoring module and said traffic clearance module are configured to operate concurrently and independently within said vehicle.

2. The system as claimed in claim 1, wherein said health monitoring module comprises:
   (a) a plurality of biomedical sensors including at least an electrocardiogram (ECG) sensor and a pulse oximeter;
   (b) a first microcontroller operatively interfaced with said plurality of biomedical sensors;
   (c) a first wireless communication module configured to transmit data from said first microcontroller to a remote server.

3. The system as claimed in claim 2, wherein said plurality of biomedical sensors further includes a blood pressure sensor and a galvanic skin response (GSR) sensor.

4. The system as claimed in claim 2, wherein said first microcontroller is configured to execute digital signal processing algorithms on data acquired from said ECG sensor, including a bandpass filter for removing baseline wander and powerline interference, and said first wireless communication module is a Wi-Fi module configured to format said physiological data into JavaScript Object Notation (JSON) formatted packets for transmission to a cloud-based medical dashboard.

5. The system as claimed in claim 1, wherein said traffic clearance module comprises:

(a) a location determining module for generating real-time geospatial coordinates of said vehicle;

(b) a second microcontroller configured to execute a geofencing algorithm using said geospatial coordinates, said algorithm computing the great-circle distance between said vehicle's current coordinates and pre-stored coordinates of traffic junctions using the Haversine formula, and triggering a clearance routine when said distance falls below a predetermined threshold;

(c) a first radio frequency (nRF) transceiver, operatively controlled by said second microcontroller, configured to transmit a digital command packet containing a junction identifier and a direction code when said geofencing algorithm detects proximity to a traffic junction.

6. The system as claimed in claim 5, further comprising a junction control unit configured for installation at a traffic intersection, said junction control unit comprising:

(a) a second nRF transceiver configured to receive said digital command packet;

(b) a third microcontroller configured to validate said received digital command packet;

(c) an interface circuit configured to override normal operation of traffic signal lights based on said validated digital command packet by executing an override sequence that: (i) first initiates a clearance phase for conflicting traffic directions; (ii) then grants a green signal to the direction specified in said digital command packet for a predetermined duration; (iii) thereafter reverts to normal traffic signal operation.

7. The system as claimed in claim 5, wherein said first nRF transceiver is an nRF24L01+ module operating in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band.

8. The system as claimed in claim 2, wherein said health monitoring module further comprises a local organic light-emitting diode (OLED) display configured to show real-time physiological parameters.

9. The system as claimed in claim 1, wherein said common power supply unit is configured to receive 12V DC input from said vehicle's electrical system and provide regulated 5V and 3.3V DC outputs to said health monitoring module and said traffic clearance module.

10. A method for integrated emergency response, comprising the concurrent steps of:

(a) continuously acquiring and remotely transmitting physiological data of a patient from a moving emergency vehicle to a medical facility;

(b) automatically detecting when said emergency vehicle approaches within a predetermined distance of a traffic junction;

(c) wirelessly transmitting a priority command from said emergency vehicle to said traffic junction;

(d) overriding normal traffic signal operation at said traffic junction based on said priority command to grant right-of-way to said emergency vehicle.

ABSTRACT

This invention is a smart system for ambulances that does two vital jobs at once: monitoring the patient and clearing traffic. The health part uses sensors (for ECG, oxygen, blood pressure, and stress) connected to a microcontroller (like Arduino Mega) to collect vital signs. It cleans up the signals, packages the data, and sends it via Wi-Fi to a hospital cloud page for doctors to see in real-time. At the same time, the traffic part uses a separate microcontroller (like Arduino Nano) with GPS. When the ambulance gets near a traffic light, it calculates the distance and automatically sends a radio command (using an nRF24L01+ chip) to that junction. A receiver box at the light gets the command, checks it, and safely overrides the normal lights to turn green for the ambulance, creating a clear path. Both parts work independently off the ambulance's 12V power, cutting down both diagnosis time and travel time to save lives.